Credit Risk Intelligence Platform

NeoStats AI Engineer Assignment — Submission Deck (auto-generated baseline)

Replace or extend this PDF with your own branded slides and UI screenshots.

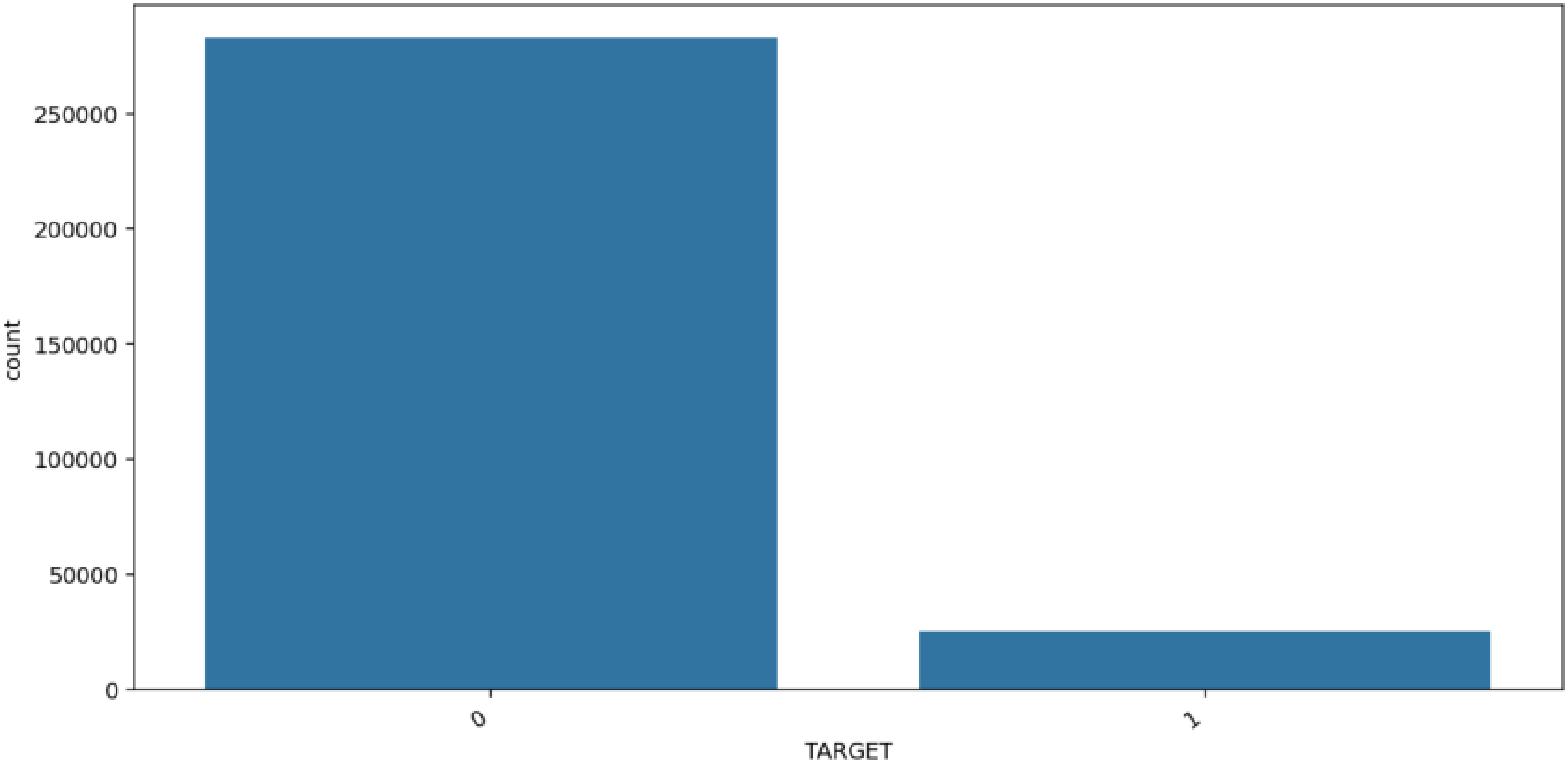
Architecture: Home Credit data → feature store → stacking ensemble → SHAP/LIME
→ SQLite → Talk-to-Data (Gemini or offline fallback) → FastAPI dashboard → Docker

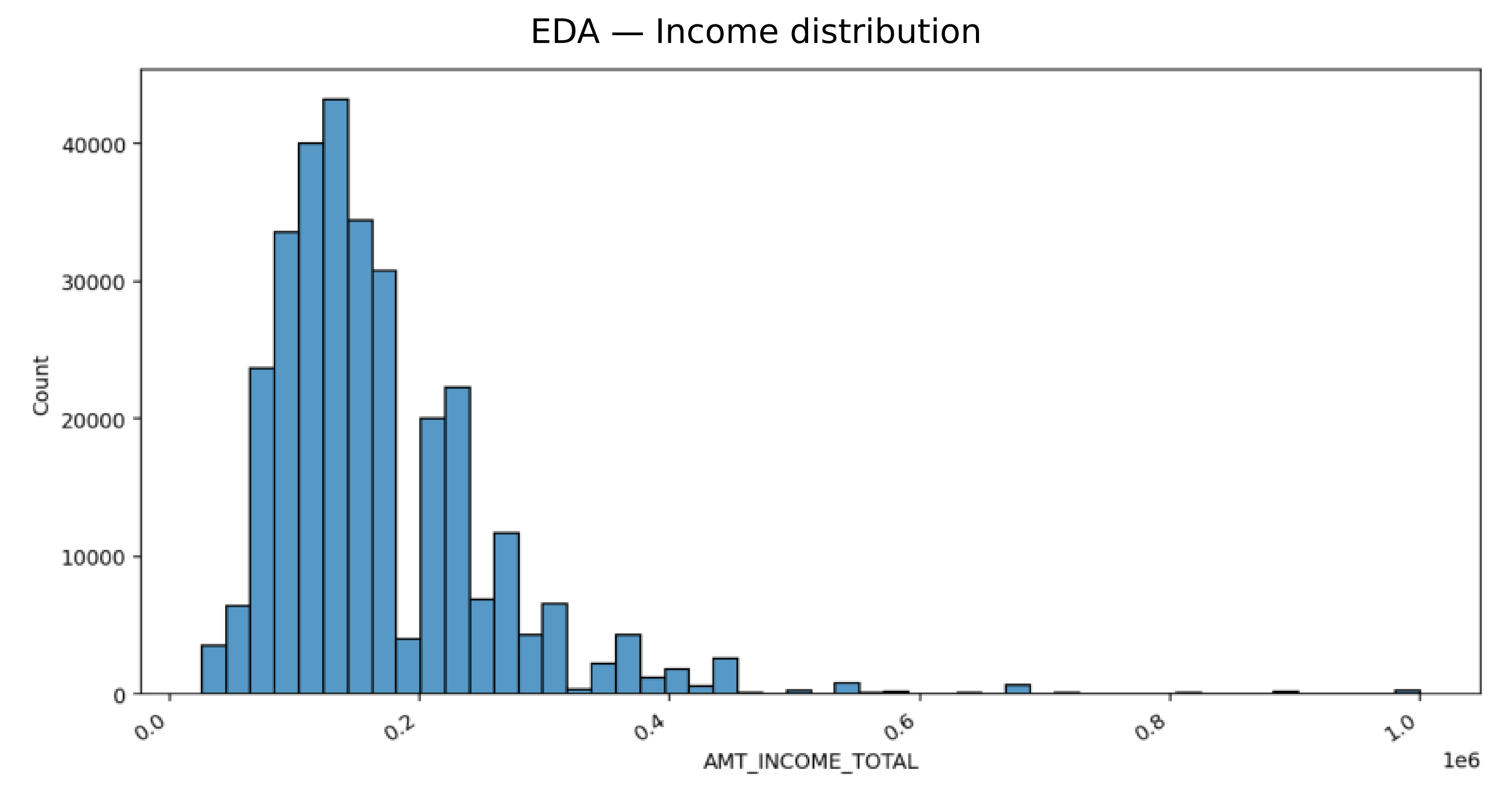
Applicants analyzed (EDA): 307511
Engineered feature categories: 11

Test metrics (stacking ensemble):
  ROC-AUC: 0.7861387631991986
  PR-AUC: 0.2863081428095381
  Recall: 0.8825780463242698
  F1:   0.22061674008810572

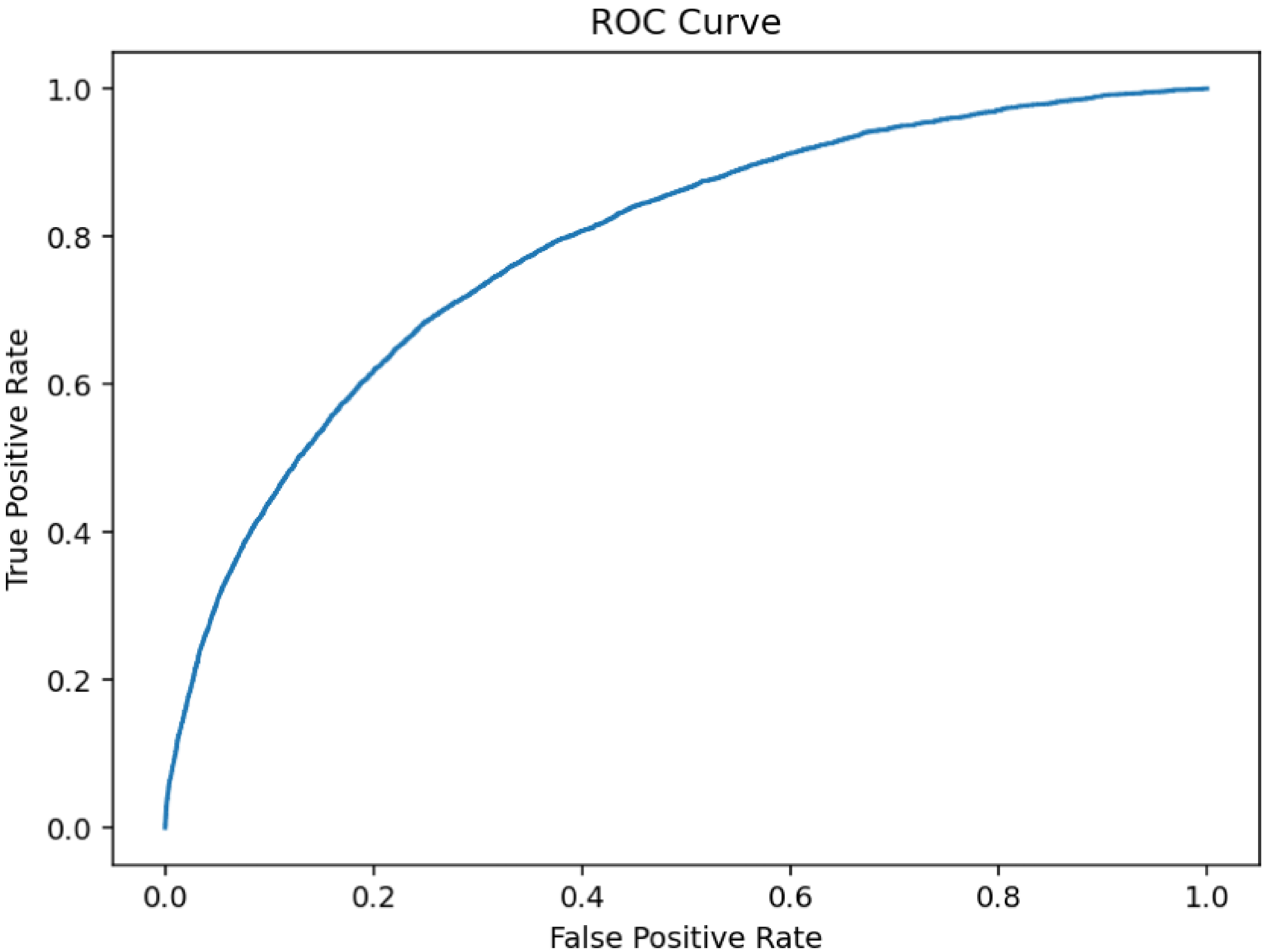
Risk bands: LOW < 0.30 | MEDIUM 0.30–0.60 | HIGH > 0.60

EDA — Target distribution

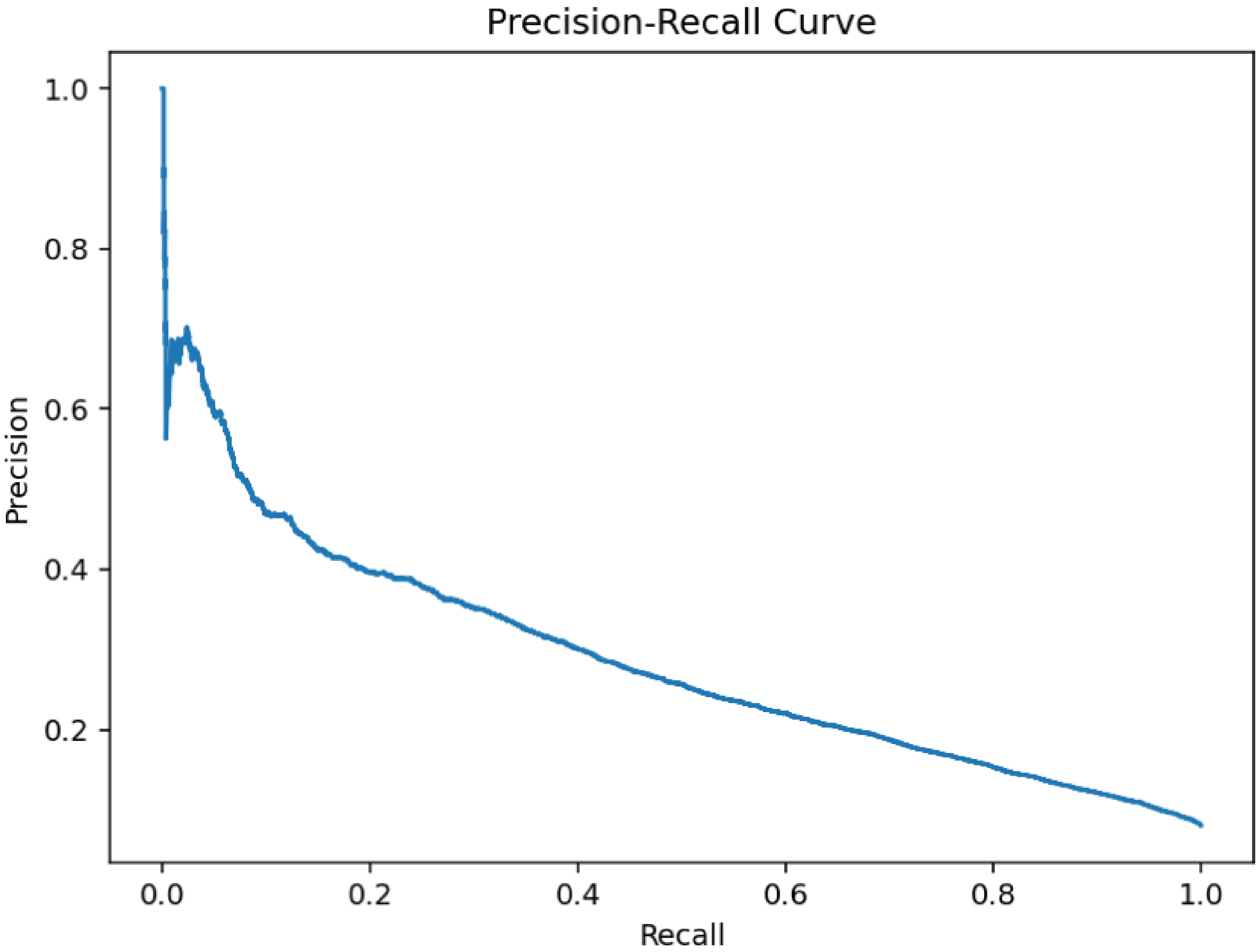


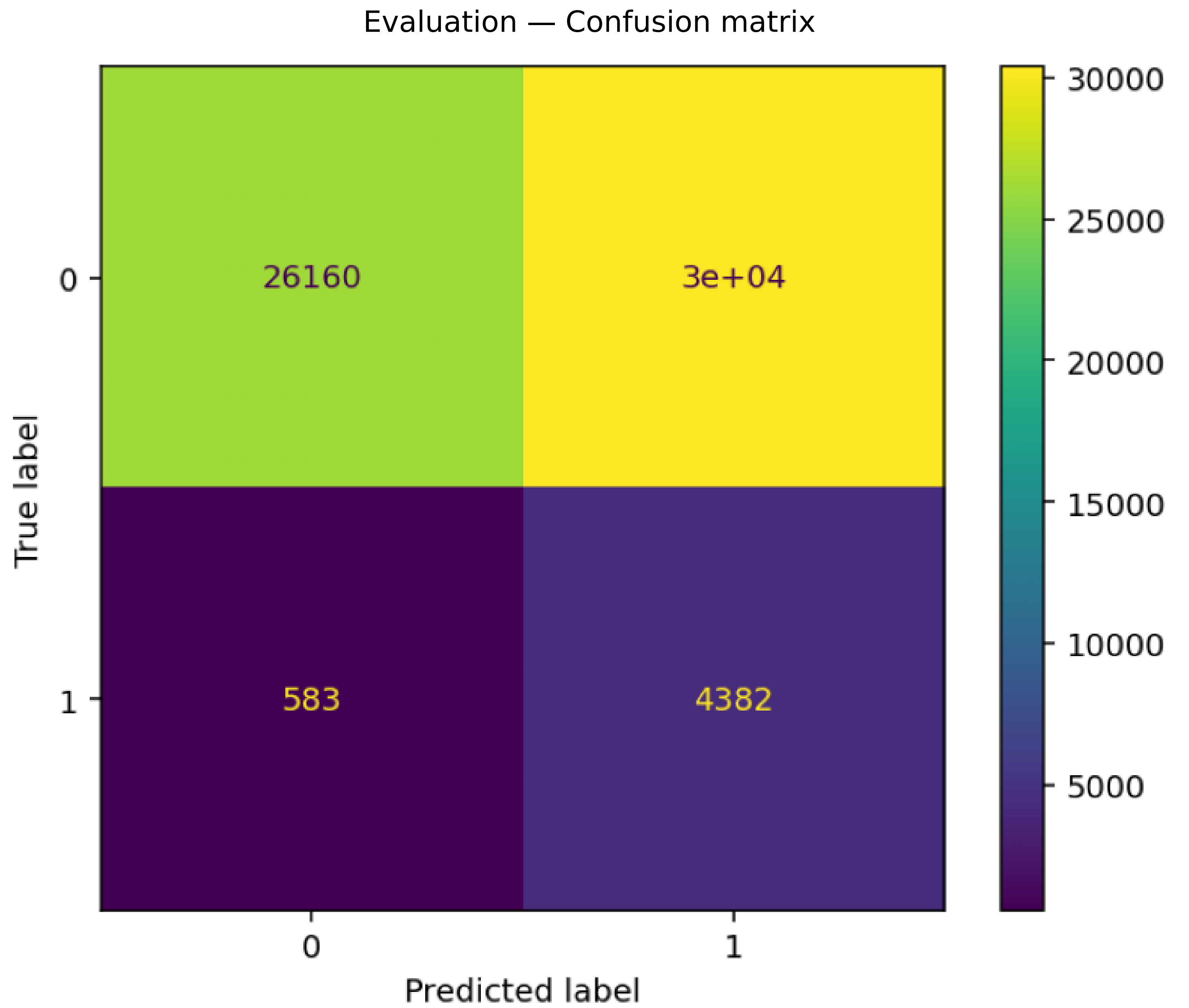

EDA — Income distribution
Count
AMT_INCOME_TOTAL
1e6
0
10000
20000
30000
40000
0.0
0.2
0.4
0.6
0.8
1.0

Evaluation — ROC curve



Evaluation — PR curve





Evaluation — Confusion matrix
26160
3e+04
583
4382
0
1
True label
0
1
Predicted label
30000
25000
20000
15000
10000
5000


SHAP — Global summary
EXT_SOURCE_2
EXT_SOURCE_3
EXT_SOURCE_1
CODE_GENDER_F
APP_GOODS_TO_CREDIT_RATIO
BUREAU_BUREAU_DEBT_TO_CREDIT_RATIO_MAX
AMT_ANNUITY
APP_EMPLOYMENT_YEARS
NAME_EDUCATION_TYPE_Higher education
INSTALLMENT_INSTALLMENT_LATE_PAYMENT_FLAG_STD
NAME_FAMILY_STATUS_Married
PREVIOUS_PREV_CREDIT_TO_APPLICATION_RATIO_MEAN
INSTALLMENT_AMT_PAYMENT_SUM
PREVIOUS_PREV_REJECTED_FLAG_MEAN
INSTALLMENT_INSTALLMENT_LATE_PAYMENT_FLAG_MEAN
AMT_GOODS_PRICE
POS_CNT_INSTALMENT_FUTURE_STD
BUREAU_BUREAU_DEBT_TO_CREDIT_RATIO_STD
DAYS_BIRTH
DAYS_EMPLOYED
0
1
SHAP value (impact on model output)
High
Low
Feature value

Talk-to-Data & Explainability

Talk-to-Data: NL → validated SQL → rows → business insight

- Gemini mode when GEMINI_API_KEY is set
- Offline fallback for 7 verified catalog questions

Explainability:

- Stacking SHAP (meta-learner on base model probabilities)
- Feature SHAP (XGBoost) + official shap.plots.waterfall PNG
- LIME (stacking black-box)

Run: uvicorn src.api.main:app --port 8000
Docker: docker-compose up --build